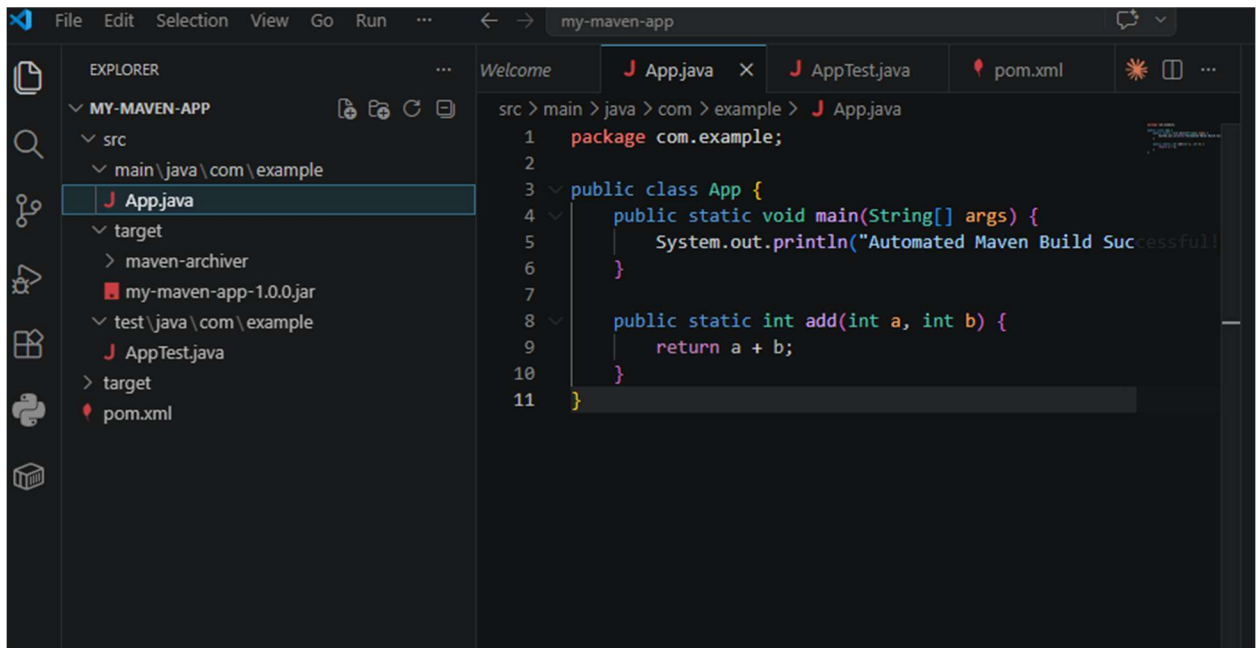
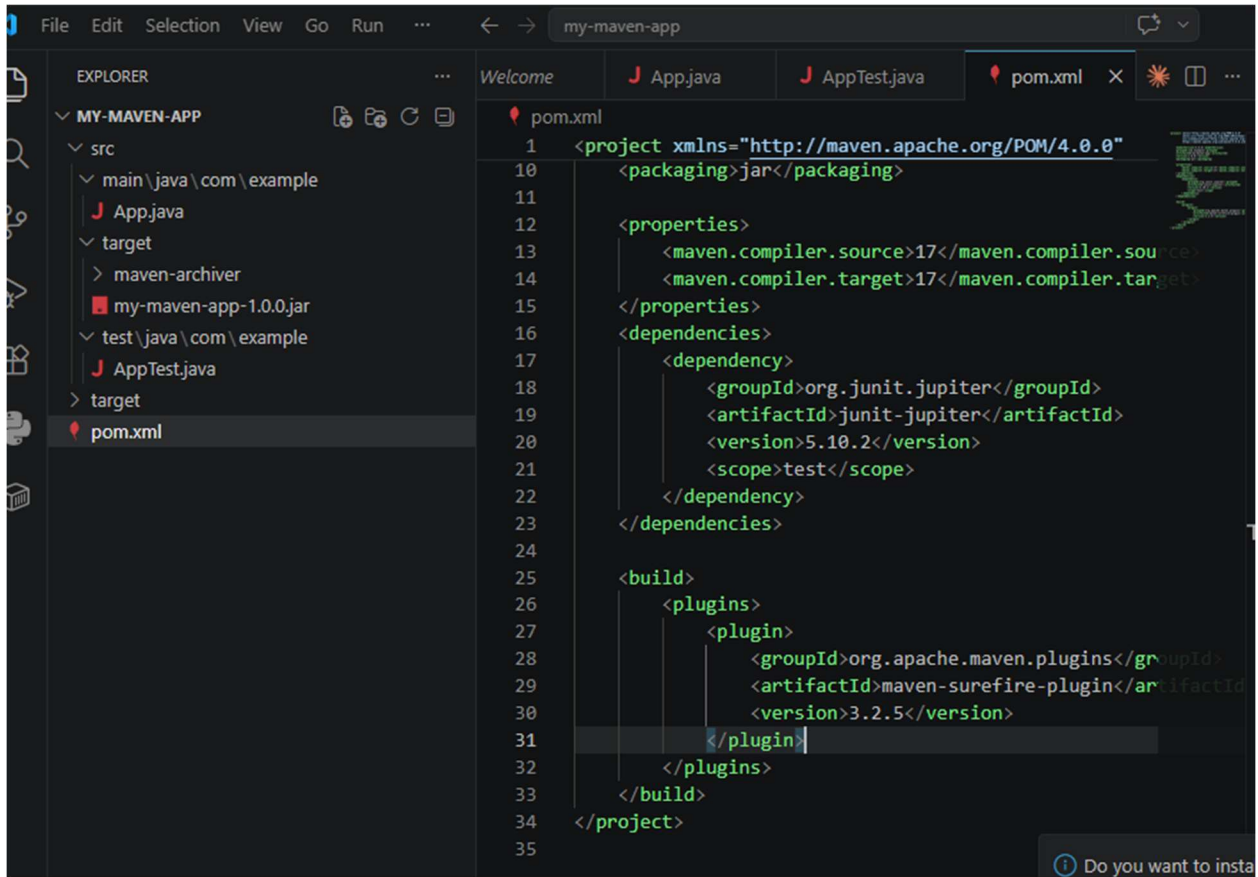
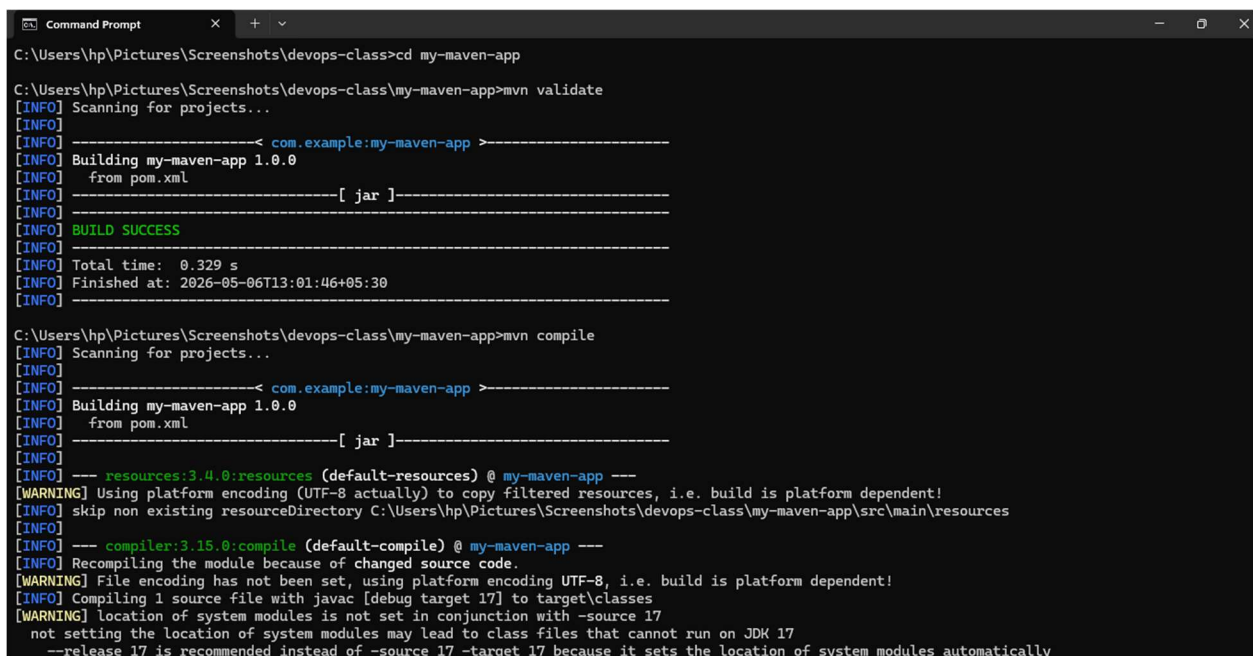
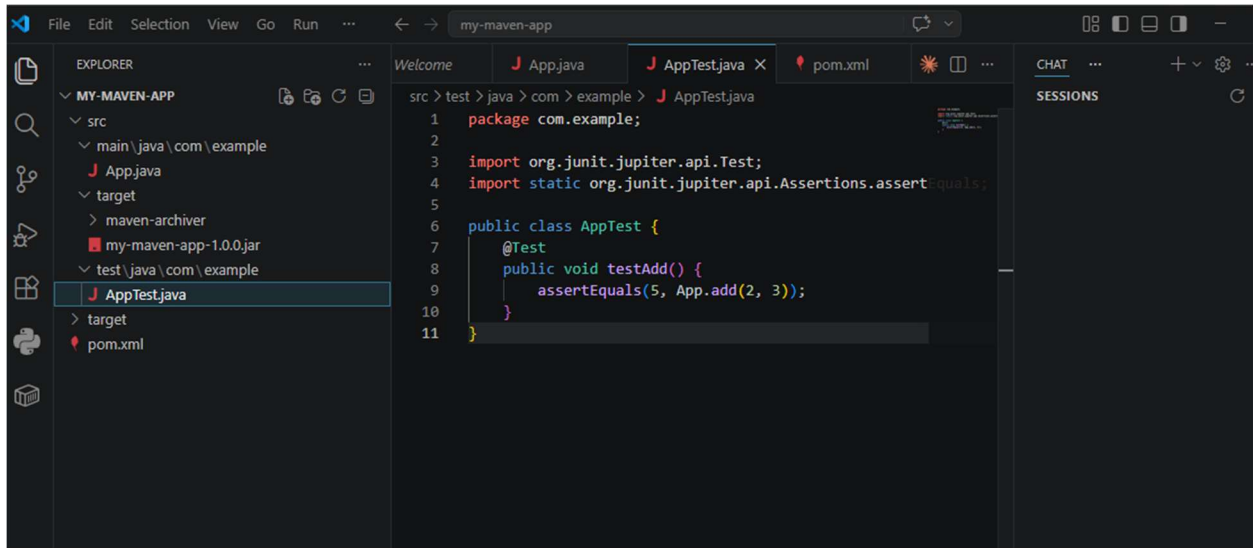


Scenario 1





```
Command Prompt
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.791 s
[INFO] Finished at: 2026-05-06T13:02:00+05:30
[INFO] -----

C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app>mvn test
[INFO] Scanning for projects...
[INFO] -----< com.example:my-maven-app >-----
[INFO] Building my-maven-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- resources:3.4.0:resources (default-resources) @ my-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app\src\main\resources
[INFO] --- compiler:3.15.0:compile (default-compile) @ my-maven-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.4.0:testResources (default-testResources) @ my-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app\src\test\resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ my-maven-app ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO] --- surefire:3.2.5:test (default-test) @ my-maven-app ---
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO] --- surefire:3.2.5:test (default-test) @ my-maven-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.122 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.924 s
[INFO] Finished at: 2026-05-06T13:02:19+05:30
[INFO] -----
```

```

C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app>mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:my-maven-app >-----
[INFO] Building my-maven-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ my-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ my-maven-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ my-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ my-maven-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ my-maven-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.215 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]

```

```

[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ my-maven-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ my-maven-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.215 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ my-maven-app ---
[INFO] Building jar: C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app\target\my-maven-app-1.0.0.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 8.696 s
[INFO] Finished at: 2026-05-06T13:04:35+05:30
[INFO] -----

C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app>java -cp target\my-maven-app-1.0.0.jar com.example.App
Automated Maven Build Successful!

```



```

C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app>mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:my-maven-app >-----
[INFO] Building my-maven-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ my-maven-app ---
[INFO] Deleting C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ my-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ my-maven-app ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] location of system modules is not set in conjunction with -source 17
    not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ my-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ my-maven-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
    not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically

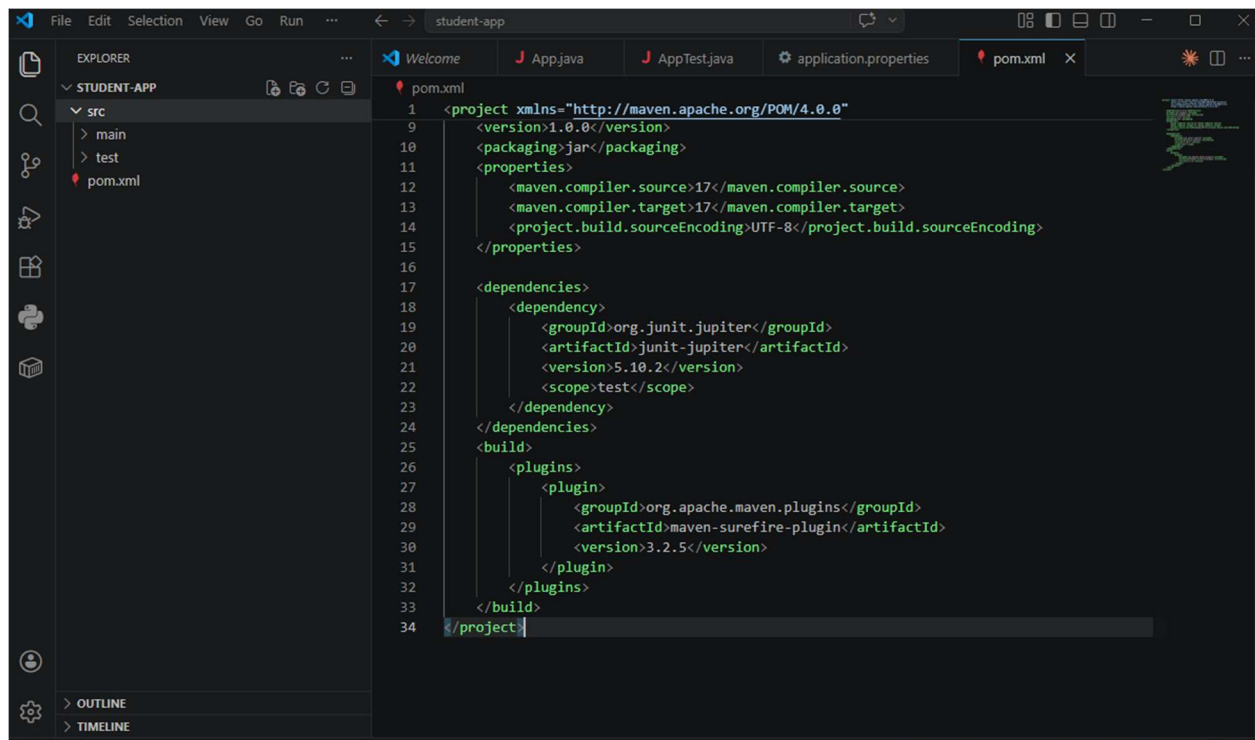
```

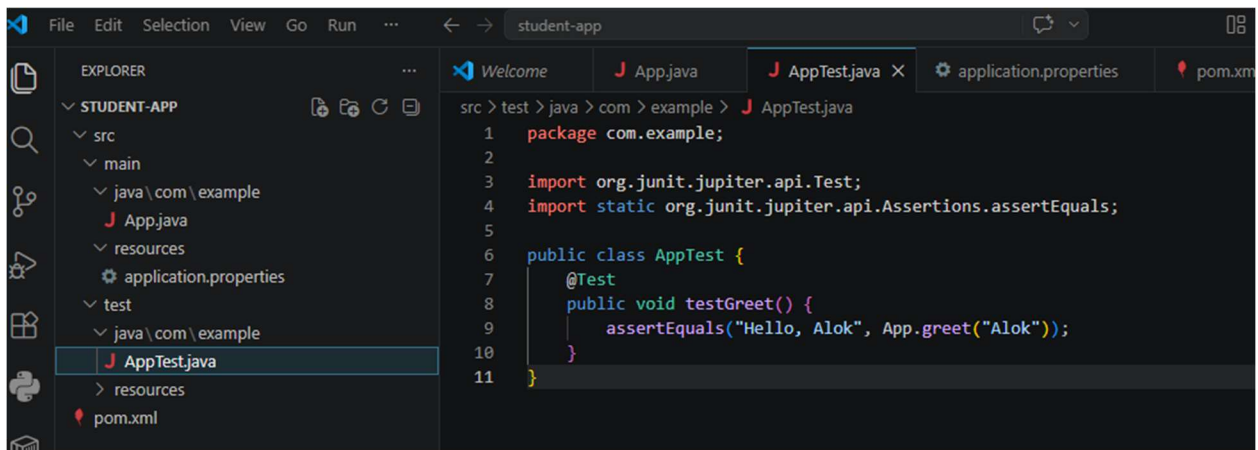
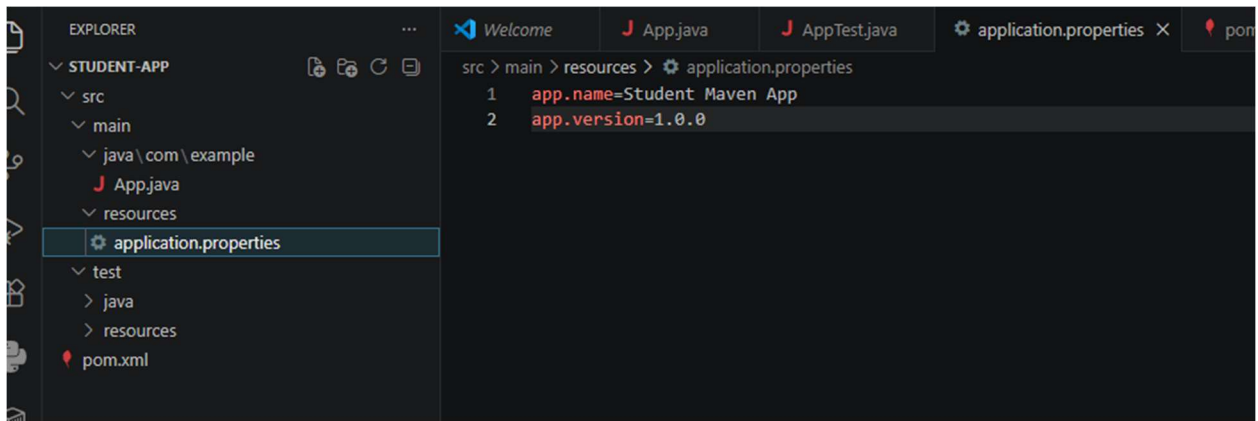
```

[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ my-maven-app ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
    not setting the location of system modules may lead to class files that cannot run on JDK 17
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ my-maven-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.122 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ my-maven-app ---
[INFO] Building jar: C:\Users\hp\Pictures\Screenshots\devops-class\my-maven-app\target\my-maven-app-1.0.0.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 7.581 s
[INFO] Finished at: 2026-05-06T13:05:48+05:30
[INFO]

```

Scenario 2





```
C:\Users\hp\Pictures\Screenshots\devops-class>mkdir student-app
C:\Users\hp\Pictures\Screenshots\devops-class>cd student-app
C:\Users\hp\Pictures\Screenshots\devops-class\student-app>mkdir src\main\java\com\example
C:\Users\hp\Pictures\Screenshots\devops-class\student-app>mkdir src\main\resources
C:\Users\hp\Pictures\Screenshots\devops-class\student-app>mkdir src\test\java\com\example
C:\Users\hp\Pictures\Screenshots\devops-class\student-app>mkdir src\test\resources
C:\Users\hp\Pictures\Screenshots\devops-class\student-app>mvn validate
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:student-app >-----
[INFO] Building student-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.270 s
[INFO] Finished at: 2026-05-06T13:16:32+05:30
[INFO]
```

```

C:\Users\hp\Pictures\Screenshots\devops-class\student-app>mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:student-app >-----
[INFO] Building student-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ student-app ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ student-app ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 1 source file with javac [debug target 17] to target/classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 2.474 s
[INFO] Finished at: 2026-05-06T13:18:29+05:30
[INFO]
C:\Users\hp\Pictures\Screenshots\devops-class\student-app>

```

```

Command Prompt
C:\Users\hp\Pictures\Screenshots\devops-class\student-app>mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:student-app >-----
[INFO] Building student-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ student-app ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ student-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ student-app ---
[INFO] Copying 0 resource from src/test/resources to target/test-classes
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ student-app ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 1 source file with javac [debug target 17] to target/test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ student-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.136 s -- in com.example.AppTest
[INFO]
[INFO] Results:

```

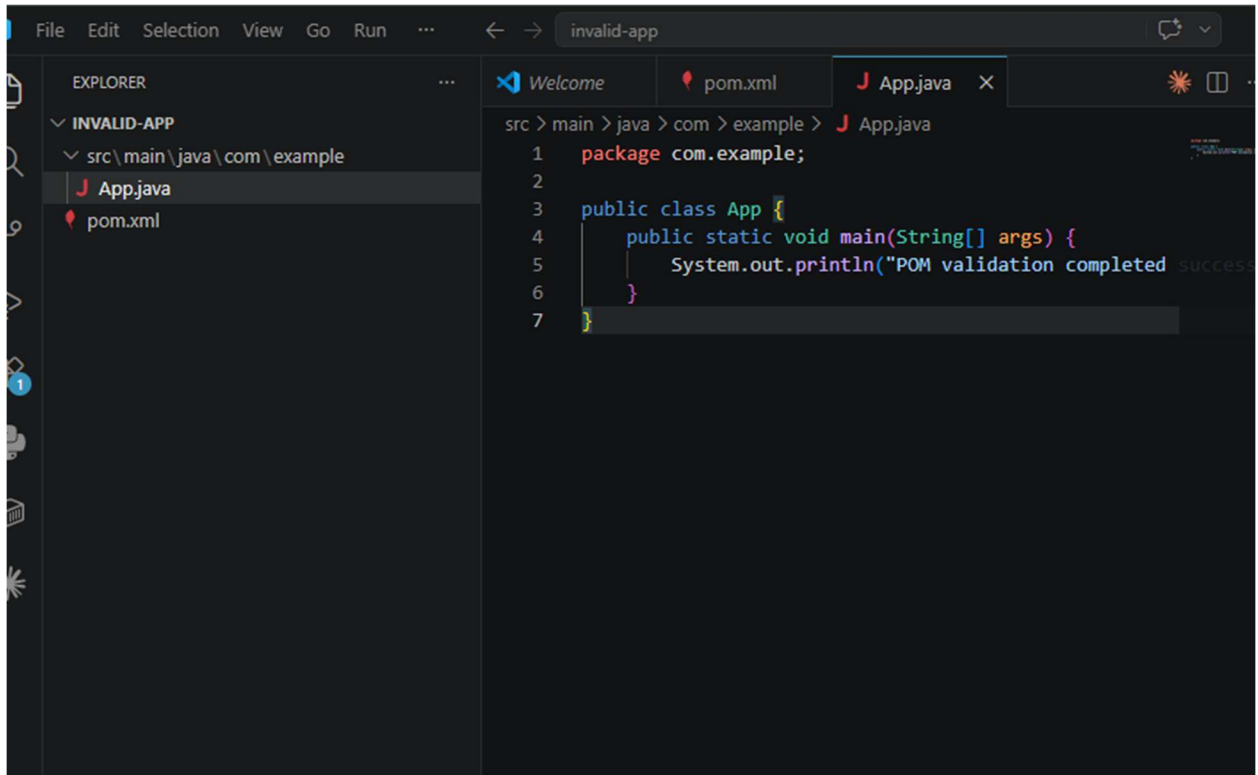


```
Command Prompt
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 5.515 s
[INFO] Finished at: 2026-05-06T13:19:07+05:30
[INFO]
[INFO] -----
C:\Users\hp\Pictures\Screenshots\devops-class\student-app>mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:student-app >-----
[INFO] Building student-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ student-app ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ student-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ student-app ---
[INFO] Copying 0 resource from src/test/resources to target/test-classes
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ student-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ student-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO]
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ student-app ---
[INFO] Copying 0 resource from src/test/resources to target/test-classes
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ student-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ student-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.183 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ student-app ---
[INFO] Building jar: C:\Users\hp\Pictures\Screenshots\devops-class\student-app\target\student-app-1.0.0.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 7.253 s
[INFO] Finished at: 2026-05-06T13:19:21+05:30
[INFO]
[INFO] -----
C:\Users\hp\Pictures\Screenshots\devops-class\student-app>java -cp target/student-app-1.0.0.jar com.example.App
Maven directory structure is correct.
C:\Users\hp\Pictures\Screenshots\devops-class\student-app>
```

```
Command Prompt
C:\Users\hp\Pictures\Screenshots\devops-class\student-app>mvn clean verify
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:student-app >-----
[INFO] Building student-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ student-app ---
[INFO] Deleting C:\Users\hp\Pictures\Screenshots\devops-class\student-app\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ student-app ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ student-app ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ student-app ---
[INFO] Copying 0 resource from src\test\resources to target\test-classes
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ student-app ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ student-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
```

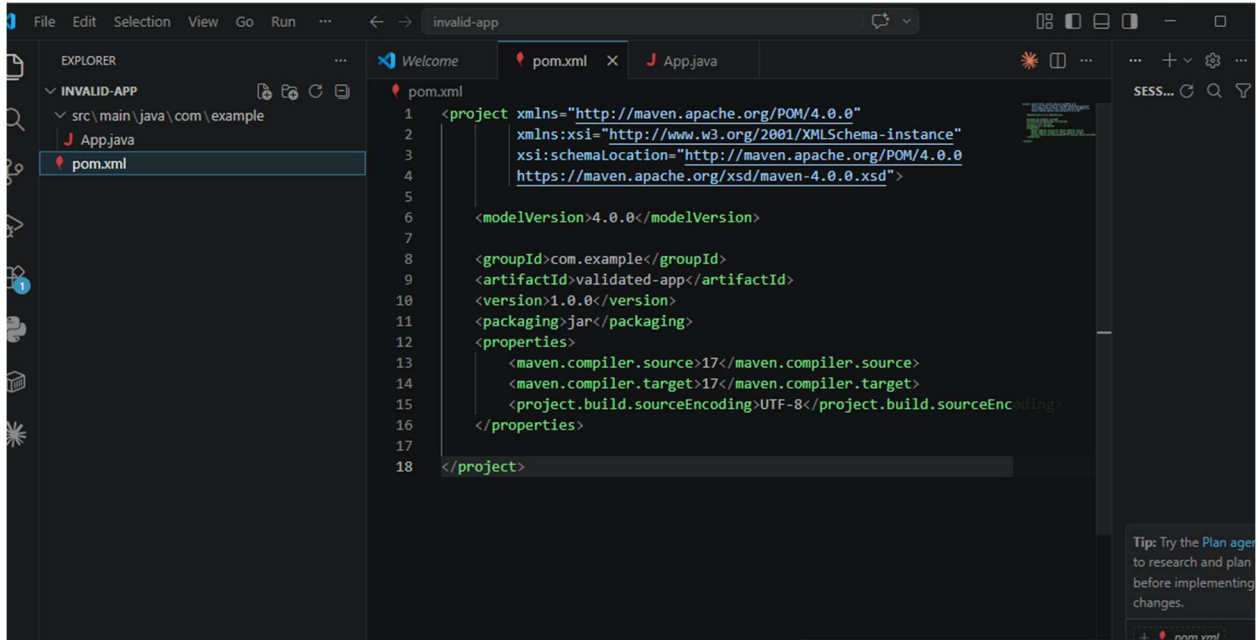
```
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ student-app ---
[INFO] Copying 0 resource from src\test\resources to target\test-classes
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ student-app ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ student-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.134 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ student-app ---
[INFO] Building jar: C:\Users\hp\Pictures\Screenshots\devops-class\student-app\target\student-app-1.0.0.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 6.795 s
[INFO] Finished at: 2026-05-06T13:21:02+05:30
[INFO]
```

Scenario 3



The screenshot shows an IDE window titled 'invalid-app'. The Explorer panel on the left shows a project structure with a folder 'INVALID-APP' containing a subfolder 'src/main/java/com/example' with files 'App.java' and 'pom.xml'. The main editor displays 'App.java' with the following code:

```
src > main > java > com > example > J App.java
1 package com.example;
2
3 public class App {
4     public static void main(String[] args) {
5         System.out.println("POM validation completed success");
6     }
7 }
```



The screenshot shows the same IDE window with 'pom.xml' selected in the Explorer and the main editor. The code is as follows:

```
1 <project xmlns="http://maven.apache.org/POM/4.0.0"
2     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3     xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4         http://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6     <modelVersion>4.0.0</modelVersion>
7
8     <groupId>com.example</groupId>
9     <artifactId>validated-app</artifactId>
10    <version>1.0.0</version>
11    <packaging>jar</packaging>
12    <properties>
13        <maven.compiler.source>17</maven.compiler.source>
14        <maven.compiler.target>17</maven.compiler.target>
15        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
16    </properties>
17
18 </project>
```

A tip box in the bottom right corner reads: "Tip: Try the Plan agent to research and plan before implementing changes."

```
C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app>mvn validate
```

```
[INFO] Scanning for projects...
```

```
[INFO]
```

```
[INFO] -----< com.example:validated-app >-----
```

```
[INFO] Building validated-app 1.0.0
```

```
[INFO] from pom.xml
```

```
[INFO] -----[ jar ]-----
```

```
[INFO]
```

```
[INFO] BUILD SUCCESS
```

```
[INFO]
```

```
[INFO] Total time: 0.231 s
```

```
[INFO] Finished at: 2026-05-06T12:20:44+05:30
```

```
[INFO]
```

```
C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app>mvn compile
```

```
[INFO] Scanning for projects...
```

```
[INFO]
```

```
[INFO] -----< com.example:validated-app >-----
```

```
[INFO] Building validated-app 1.0.0
```

```
[INFO] from pom.xml
```

```
[INFO] -----[ jar ]-----
```

```
[INFO]
```

```
[INFO] --- resources:3.4.0:resources (default-resources) @ validated-app ---
```

```
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app\src\main\resources
```

```
[INFO]
```

```
[INFO] --- compiler:3.15.0:compile (default-compile) @ validated-app ---
```

```
[INFO] Recompiling the module because of changed source code.
```

```
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
```

```
[WARNING] location of system modules is not set in conjunction with -source 17
```

```
not setting the location of system modules may lead to class files that cannot run on JDK 17
```

```
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
```

```
[INFO]
```

```
[INFO] BUILD SUCCESS
```

```
[INFO]
```

```
[INFO] Total time: 2.890 s
```

```
[INFO] Finished at: 2026-05-06T12:22:43+05:30
```

```
[INFO]
```

```
C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app>
```

```
C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app>mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:validated-app >-----
[INFO] Building validated-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ validated-app ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ validated-app ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ validated-app ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ validated-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ validated-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ validated-app ---
[INFO] Building jar: C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app\target\validated-app-1.0.0.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.761 s
[INFO] Finished at: 2026-05-06T12:23:32+05:30
[INFO]
```

```
Command Prompt
C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app>mvn help:effective-pom
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (14 kB at 9.8 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml (21 kB at 11 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-help-plugin/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-help-plugin/maven-metadata.xml (807 B at 12 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-help-plugin/3.5.1/maven-help-plugin-3.5.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-help-plugin/3.5.1/maven-help-plugin-3.5.1.pom (10 kB at 141 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/43/maven-plugins-43.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/43/maven-plugins-43.pom (7.5 kB at 100 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-help-plugin/3.5.1/maven-help-plugin-3.5.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-help-plugin/3.5.1/maven-help-plugin-3.5.1.jar (65 kB at 938 kB/s)
[INFO]
[INFO] -----< com.example:validated-app >-----
[INFO] Building validated-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- help:3.5.1:effective-pom (default-cli) @ validated-app ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugin-tools/maven-plugin-tools-generators/3.13.1/maven-plugin-tools-generators-3.13.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugin-tools/maven-plugin-tools-generators/3.13.1/maven-plugin-tools-generators-3.13.1.pom (4.1 kB at 9.8 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugin-tools/maven-plugin-tools/3.13.1/maven-plugin-tools-3.13.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugin-tools/maven-plugin-tools/3.13.1/maven-plugin-tools-3.13.1.pom (18 kB at 287 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/sisu/org.eclipse.sisu.plexus/0.9.0.M3/org.eclipse.sisu.plexus-0.9.0.M3.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/sisu/org.eclipse.sisu.plexus/0.9.0.M3/org.eclipse.sisu.plexus-0.9.0.M3.pom (17 kB at 262 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/sisu/sisu-inject/0.9.0.M3/sisu-inject-0.9.0.M3.pom
```



```
Command Prompt

<reportPlugins>
  <reportPlugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-project-info-reports-plugin</artifactId>
  </reportPlugin>
</reportPlugins>
</configuration>
</execution>
</executions>
<configuration>
  <outputDirectory>C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app\target\site</outputDirectory>
  <reportPlugins>
    <reportPlugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-project-info-reports-plugin</artifactId>
    </reportPlugin>
  </reportPlugins>
</configuration>
</plugin>
</plugins>
</build>
<reporting>
  <outputDirectory>C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app\target\site</outputDirectory>
</reporting>
</project>

[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 6.888 s
[INFO] Finished at: 2026-05-06T12:24:21+05:30
[INFO] -----
```

```
Command Prompt

C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app>mvn -X validate
Apache Maven 3.9.15 (98b2cdebdb5f1ac8781f537ea9acccaed7922349)
Maven home: D:\apache-maven-3.9.15-bin\apache-maven-3.9.15
Java version: 26.0.1, vendor: Oracle Corporation, runtime: C:\Program Files\Java\jdk-26.0.1
Default locale: en_US, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
[DEBUG] Created new class realm maven.api
[DEBUG] Importing foreign packages into class realm maven.api
[DEBUG] Imported: javax.annotation.* < plexus.core
[DEBUG] Imported: javax.annotation.security.* < plexus.core
[DEBUG] Imported: javax.inject.* < plexus.core
[DEBUG] Imported: org.apache.maven.* < plexus.core
[DEBUG] Imported: org.apache.maven.artifact < plexus.core
[DEBUG] Imported: org.apache.maven.classrealm < plexus.core
[DEBUG] Imported: org.apache.maven.cli < plexus.core
[DEBUG] Imported: org.apache.maven.configuration < plexus.core
[DEBUG] Imported: org.apache.maven.exception < plexus.core
[DEBUG] Imported: org.apache.maven.execution < plexus.core
[DEBUG] Imported: org.apache.maven.execution.scope < plexus.core
[DEBUG] Imported: org.apache.maven.graph < plexus.core
[DEBUG] Imported: org.apache.maven.lifecycle < plexus.core
[DEBUG] Imported: org.apache.maven.model < plexus.core
[DEBUG] Imported: org.apache.maven.monitor < plexus.core
[DEBUG] Imported: org.apache.maven.plugin < plexus.core
[DEBUG] Imported: org.apache.maven.profiles < plexus.core
[DEBUG] Imported: org.apache.maven.project < plexus.core
[DEBUG] Imported: org.apache.maven.reporting < plexus.core
[DEBUG] Imported: org.apache.maven.repository < plexus.core
[DEBUG] Imported: org.apache.maven.rtinfo < plexus.core
[DEBUG] Imported: org.apache.maven.settings < plexus.core
[DEBUG] Imported: org.apache.maven.toolchain < plexus.core
[DEBUG] Imported: org.apache.maven.usability < plexus.core
[DEBUG] Imported: org.apache.maven.wagon.* < plexus.core
[DEBUG] Imported: org.apache.maven.wagon.authentication < plexus.core
```

```

Command Prompt

[DEBUG] Project: com.example:validated-app:jar:1.0.0
[DEBUG] Tasks: [validate]
[DEBUG] Style: Regular
[DEBUG] =====
[INFO] -----< com.example:validated-app >-----
[INFO] Building validated-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[DEBUG] Lifecycle clean -> [pre-clean, clean, post-clean]
[DEBUG] Lifecycle default -> [validate, initialize, generate-sources, process-sources, generate-resources, process-resources, compile, process-classes, generate-test-sources, process-test-sources, generate-test-resources, process-test-resources, test-compile, process-test-classes, test, prepare-package, package, pre-integration-test, integration-test, post-integration-test, verify, install, deploy]
[DEBUG] Lifecycle site -> [pre-site, site, post-site, site-deploy]
[DEBUG] === PROJECT BUILD PLAN ===
[DEBUG] Project: com.example:validated-app:1.0.0
[DEBUG] Dependencies (collect): []
[DEBUG] Dependencies (resolve): []
[DEBUG] Repositories (dependencies): [central (https://repo.maven.apache.org/maven2, default, releases)]
[DEBUG] Repositories (plugins) : [central (https://repo.maven.apache.org/maven2, default, releases)]
[DEBUG] =====
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.282 s
[INFO] Finished at: 2026-05-06T12:25:34+05:30
[INFO]
[DEBUG] Shutting down adapter factory; available factories [file-lock, rwlock-local, semaphore-local, noop]; available name mappers [discriminating, file-gaev, file-gav, file-hgaev, file-hgav, file-static, gaev, gav, static]
[DEBUG] Shutting down 'file-lock' factory
[DEBUG] Shutting down 'rwlock-local' factory
[DEBUG] Shutting down 'semaphore-local' factory
[DEBUG] Shutting down 'noop' factory

```

```

Command Prompt

C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app>mvn validate
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:validated-app >-----
[INFO] Building validated-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.212 s
[INFO] Finished at: 2026-05-06T12:26:23+05:30
[INFO]

C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app>mvn clean verify
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:validated-app >-----
[INFO] Building validated-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-clean-plugin-3.2.0.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-clean-plugin-3.2.0.pom (5.3 kB at 3.5 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/35/maven-plugins-35.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/35/maven-plugins-35.pom (9.9 kB at 202 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/35/maven-parent-35.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/35/maven-parent-35.pom (45 kB at 526 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/25/apache-25.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/25/apache-25.pom (21 kB at 311 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-clean-plugin-3.2.0.jar
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-clean-plugin-3.2.0.jar (36 kB at 469 kB/s)
[INFO]

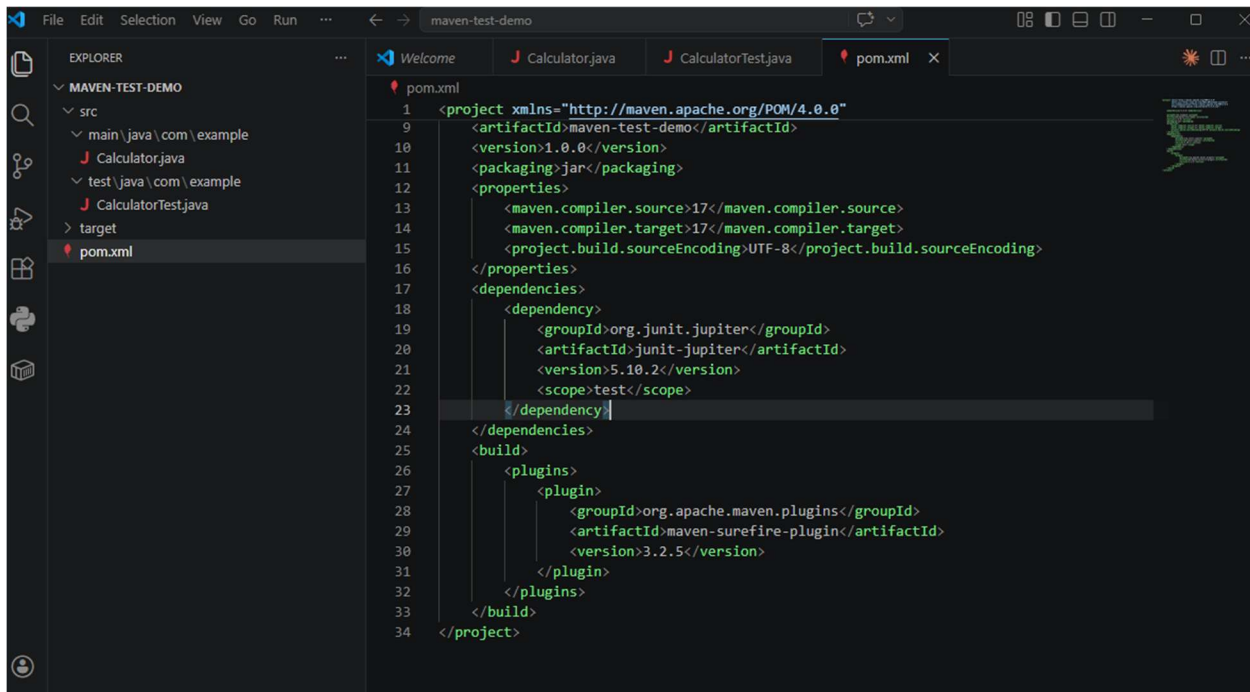
```

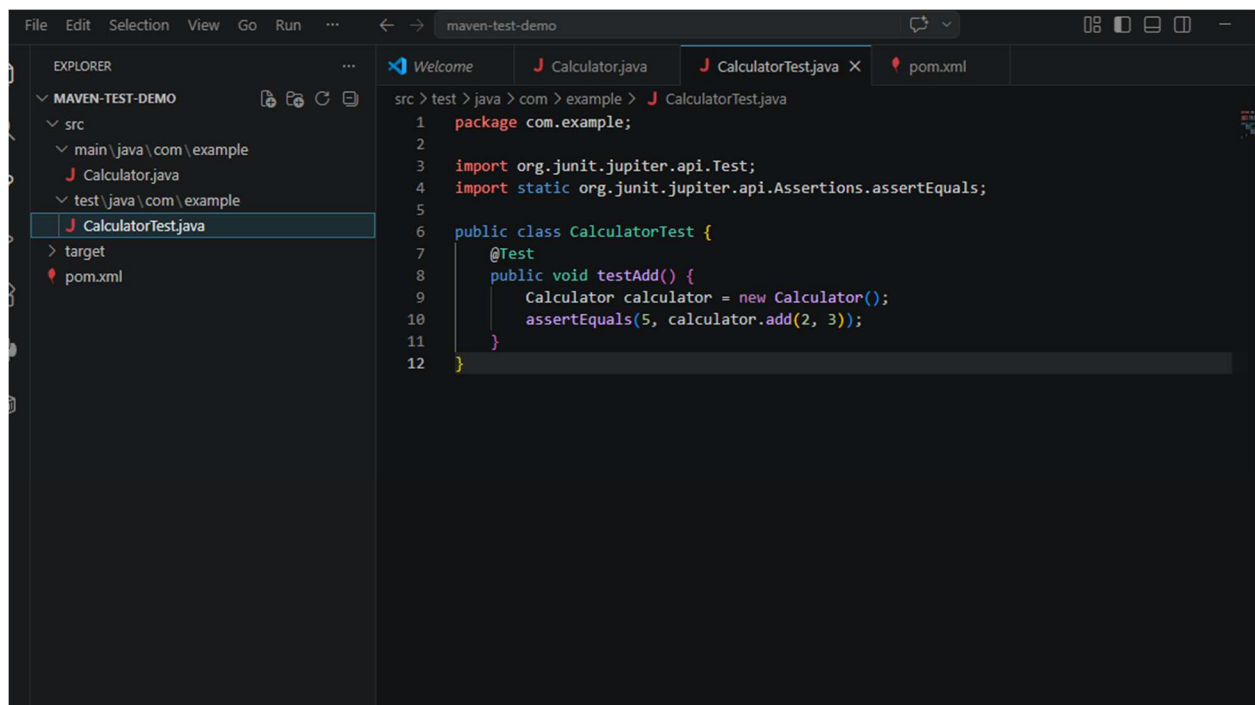
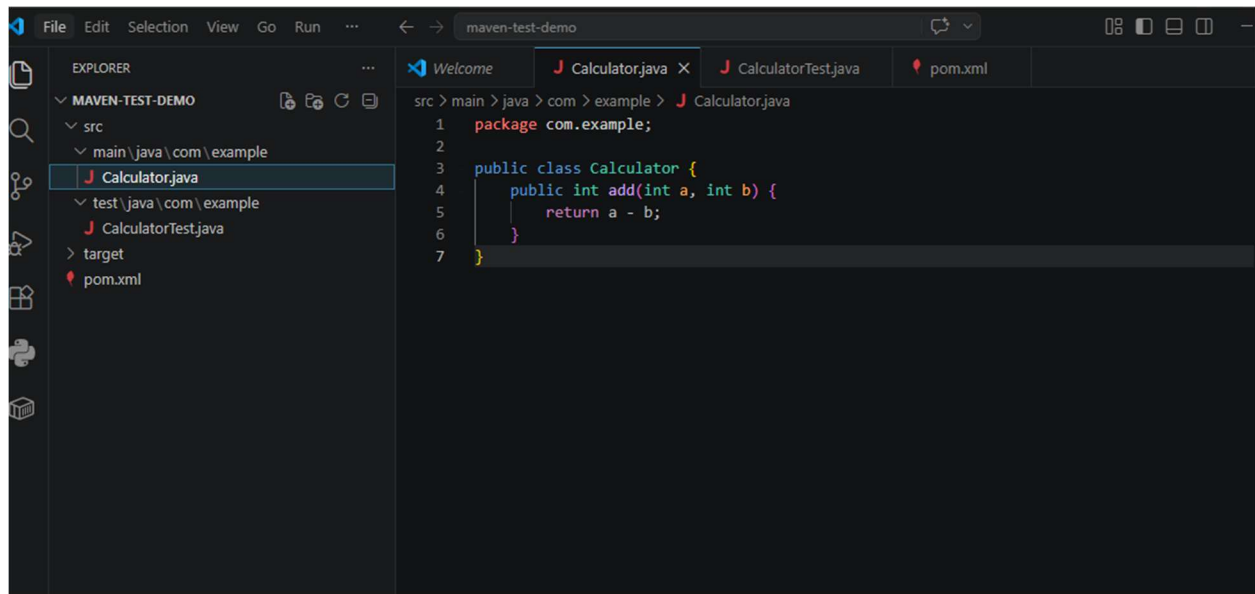
```

Command Prompt
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-utils/3.3.4/maven-shared-utils-3.3.4.jar (153 kB
at 2.0 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.6/commons-io-2.6.jar
Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.6/commons-io-2.6.jar (215 kB at 3.4 MB/s)
[INFO] Deleting C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ validated-app ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ validated-app ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ validated-app ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ validated-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ validated-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ validated-app ---
[INFO] Building jar: C:\Users\hp\Pictures\Screenshots\devops-class\invalid-app\target\validated-app-1.0.0.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 7.203 s
[INFO] Finished at: 2026-05-06T12:26:44+05:30
[INFO]

```

Scenario 4






```
C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo>mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:maven-test-demo >-----
[INFO] Building maven-test-demo 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ maven-test-demo ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ maven-test-demo ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 2.831 s
[INFO] Finished at: 2026-05-06T12:36:36+05:30
[INFO]
[INFO] -----
```

```
Command Prompt
C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo>mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:maven-test-demo >-----
[INFO] Building maven-test-demo 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ maven-test-demo ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ maven-test-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ maven-test-demo ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ maven-test-demo ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ maven-test-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit-platform/3.2.5/surefire-junit-platform-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-junit-platform/3.2.5/surefire-junit-platform-3.2.5.pom (4.7 kB at 3.8 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-providers/3.2.5/surefire-providers-3.2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-providers/3.2.5/surefire-providers-3.2.5.pom (2.6 kB at 4.9 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/common-java5/3.2.5/common-java5-3.2.5.pom
```

```
[INFO]
[INFO] Results:
[INFO]
[ERROR] Failures:
[ERROR]   CalculatorTest.testAdd:10 expected: <5> but was: <-1>
[INFO]
[ERROR] Tests run: 1, Failures: 1, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] BUILD FAILURE
[INFO]
[INFO] Total time: 13.490 s
[INFO] Finished at: 2026-05-06T12:38:37+05:30
[INFO]
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-surefire-plugin:3.2.5:test (default-test) on project maven-test-demo: There are test failures.
[ERROR]
[ERROR] Please refer to C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\target\surefire-reports for the individual test results.
[ERROR] Please refer to dump files (if any exist) [date].dump, [date]-jvmRun[N].dump and [date].dumpstream.
[ERROR] -> [Help 1]
[ERROR]
[ERROR] To see the full stack trace of the errors, re-run Maven with the -e switch.
[ERROR] Re-run Maven using the -X switch to enable full debug logging.
[ERROR]
[ERROR] For more information about the errors and possible solutions, please read the following articles:
[ERROR] [Help 1] http://cwiki.apache.org/confluence/display/MAVEN/MojoFailureException
```



```
Welcome X Calculator.java X CalculatorTest.java pom.xml
src > main > java > com > example > Calculator.java
1 package com.example;
2
3 public class Calculator {
4     public int add(int a, int b) {
5         return a + b;
6     }
7 }
```

```
Command Prompt
C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo>mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:maven-test-demo >-----
[INFO] Building maven-test-demo 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ maven-test-demo ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ maven-test-demo ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ maven-test-demo ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ maven-test-demo ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ maven-test-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
```

```
Command Prompt
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO] --- resources:3.4.0:testResources (default-testResources) @ maven-test-demo ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\src\test\resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ maven-test-demo ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO] --- surefire:3.2.5:test (default-test) @ maven-test-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.CalculatorTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.145 s -- in com.example.CalculatorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 8.508 s
[INFO] Finished at: 2026-05-06T12:40:11+05:30
[INFO]
```

```
Command Prompt
C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo>mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:maven-test-demo >-----
[INFO] Building maven-test-demo 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ maven-test-demo ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\src\main\resources
[INFO] --- compiler:3.15.0:compile (default-compile) @ maven-test-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.4.0:testResources (default-testResources) @ maven-test-demo ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\src\test\resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ maven-test-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- surefire:3.2.5:test (default-test) @ maven-test-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.CalculatorTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.139 s -- in com.example.CalculatorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

```

[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ maven-test-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ maven-test-demo ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\src\test\r
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ maven-test-demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ maven-test-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.CalculatorTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.139 s -- in com.example.CalculatorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ maven-test-demo ---
[INFO] Building jar: C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\target\maven-test-demo-1.0.0.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 5.793 s
[INFO] Finished at: 2026-05-06T12:41:14+05:30
[INFO] -----

```

```

C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo>mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:maven-test-demo >-----
[INFO] Building maven-test-demo 1.0.0
[INFO] from pom.xml
[INFO]
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ maven-test-demo ---
[INFO] Deleting C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ maven-test-demo ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ maven-test-demo ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ maven-test-demo ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ maven-test-demo ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ maven-test-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider

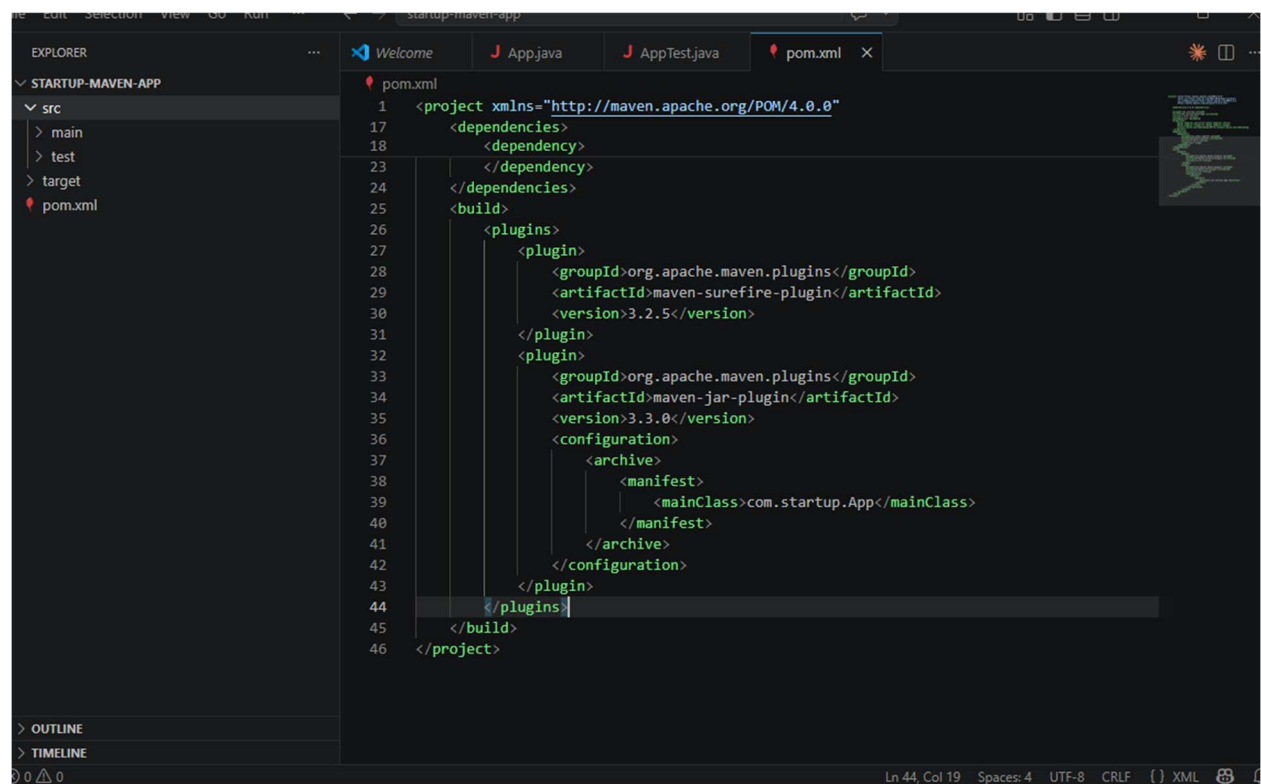
```

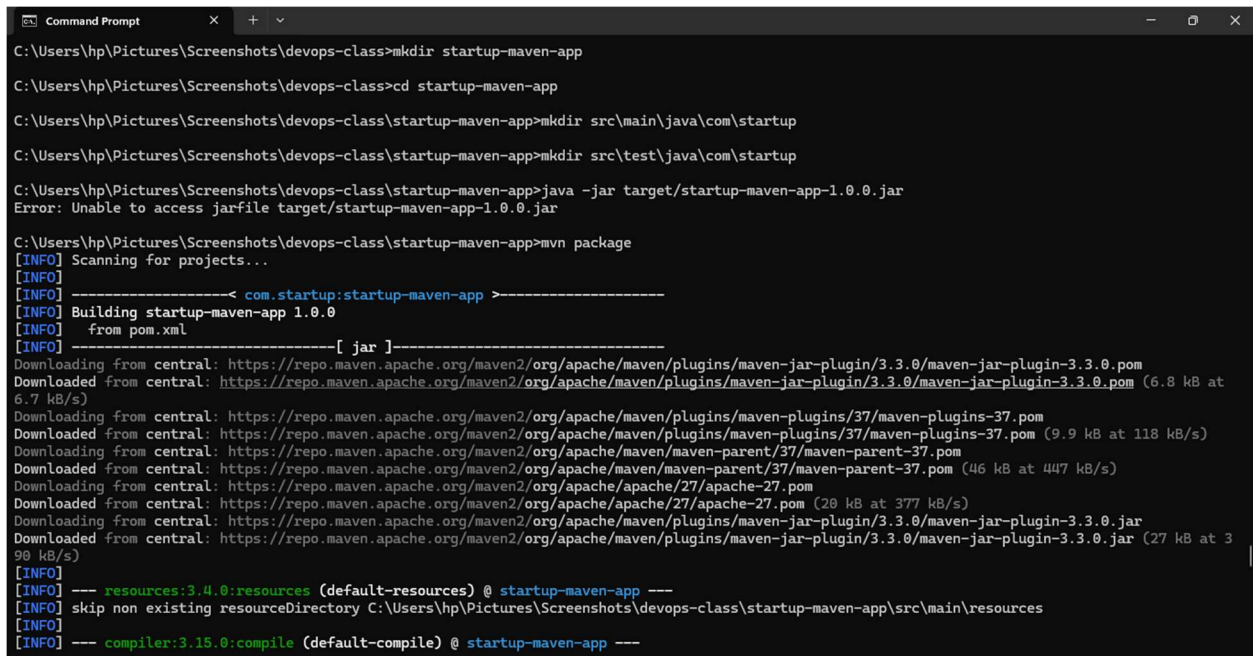
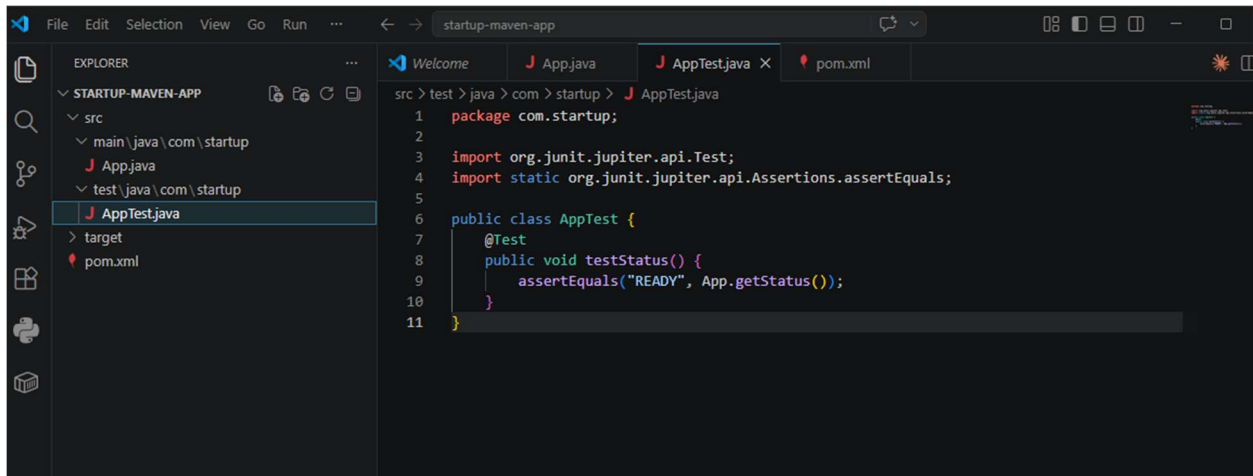
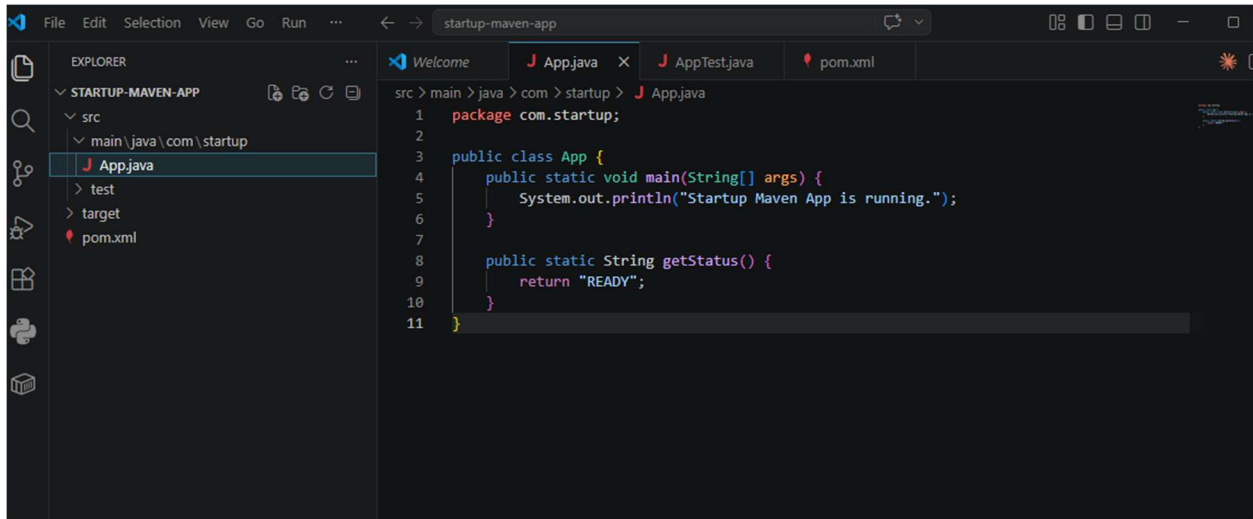
```

not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO] --- resources:3.4.0:testResources (default-testResources) @ maven-test-demo ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo\src\test\resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ maven-test-demo ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO] --- surefire:3.2.5:test (default-test) @ maven-test-demo ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.CalculatorTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.127 s -- in com.example.CalculatorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 8.134 s
[INFO] Finished at: 2026-05-06T12:42:04+05:30
[INFO]
C:\Users\hp\Pictures\Screenshots\devops-class\maven-test-demo>

```

Scenario 5






```

Command Prompt
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.4.2/plexus-utils-3.4.2.jar (267 kB at 357 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.21/commons-compress-1.21.jar (1.0 MB at 1.0 MB/s)
[INFO] Building jar: C:\Users\hp\Pictures\Screenshots\devops-class\startup-maven-app\target\startup-maven-app-1.0.0.jar
[INFO] BUILD SUCCESS
[INFO] Total time: 11.311 s
[INFO] Finished at: 2026-05-06T12:50:10+05:30

C:\Users\hp\Pictures\Screenshots\devops-class\startup-maven-app>java -jar target\startup-maven-app-1.0.0.jar
Startup Maven App is running.

C:\Users\hp\Pictures\Screenshots\devops-class\startup-maven-app>mvn clean package
[INFO] Scanning for projects...
[INFO] -----< com.startup:startup-maven-app >-----
[INFO] Building startup-maven-app 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- clean:3.2.0:clean (default-clean) @ startup-maven-app ---
[INFO] Deleting C:\Users\hp\Pictures\Screenshots\devops-class\startup-maven-app\target
[INFO] --- resources:3.4.0:resources (default-resources) @ startup-maven-app ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\startup-maven-app\src\main\resources
[INFO] --- compiler:3.15.0:compile (default-compile) @ startup-maven-app ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO]

```

```

Command Prompt
[INFO] --- resources:3.4.0:testResources (default-testResources) @ startup-maven-app ---
[INFO] skip non existing resourceDirectory C:\Users\hp\Pictures\Screenshots\devops-class\startup-maven-app\src\test\resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ startup-maven-app ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO] --- surefire:3.2.5:test (default-test) @ startup-maven-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.startup.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.149 s -- in com.startup.AppTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] --- jar:3.3.0:jar (default-jar) @ startup-maven-app ---
[INFO] Building jar: C:\Users\hp\Pictures\Screenshots\devops-class\startup-maven-app\target\startup-maven-app-1.0.0.jar
[INFO] BUILD SUCCESS
[INFO] Total time: 7.002 s
[INFO] Finished at: 2026-05-06T12:51:40+05:30
[INFO]

```